

K4 Spies Solution and Procedure

From Ibn Şifre to Yahoo Kryptos Group:

Bismillah, Let us begin:

Part 1: K4-H

Carryout Transpositions Step:

Start with K3 and arrange it in 7 rows by 48 columns as shown in Fig 1-1.

	1	2	3	4
	123456789012345678901234567890123456789012345678			
1	ENDYAHROHNLSRHEOCPTEOIBIDYSHNAIACHTNREYULDSL	LSLL		
2	NOHSNOSMXPNETPRNGATIHNRARPESLNNELEBLPIIACA	EWMT		
3	WNDITEENRAHCTENEUDRETNHAEOETFOLSEDTIWENHAEIOY	TEY		
4	QHEENCTAYCREIFTBRSPAMHHEWENATAMATEGYEERLBTEE	FOAS		
5	FIOTUETUAEOTOARMAEERTNRTIBSEDDNIAAHTTMSTEWPI	EROA		
6	GRIEWFEBAECTDDHILCEIHSITEGOEAOSSDDRYDLORITRKL	MLEH		
7	AGTDHARDPNEOHMGFMFEUHEECDM RIPFEIMEHNLSSTTRTV	DOHW		

Fig. 1-1, K3

Flip K3 about the columns so that column 48 is first and column 1 is last, as shown in Fig 1-2. This is done only to make the techniques easier to visualize. Call it K3'.

	4	3	2	1
	876543210987654321098765432109876543210987654321			
1	LLSLLSDLUYERNTHCAIANHSYDIBIOETPCOEHRSLNHORHAYDNE			
2	TMWEACAIIPLBELENNLSEPRARNHITAGNRPTENMXWRMSONSHON			
3	YETYOIEAHNEWITDESLOFTEOEAHNTERDUENETCHARNEETIDNW			
4	SAOFEETBLREEYGETAMATANEWEEHMAPSRBTFIERCYATCNEEHQ			
5	AOREIPWETSMTTHAAINDDESBITRNTREEAMRAOTOE AUTEUTOIF			
6	HEMLKRTIROLDYRDDSOAE OGETISHIECLIHDDTCEABEFWEIRG			
7	WHODVTRTTSSLNHEMIEFFIRMDCEEHUEFMFGMHOENPDRAHDTGA			

Fig. 1-2, K3'

Using the END - YAR (RAY) clue, place K3' in the form of a transverse EM wave with a period of 16 characters and an amplitude of 11 characters, as shown in Fig 1-3.

	4	3	2	1		
	87654321	0987654321	0987654321	0987654321	0987654321	
4	LLSD	RNTH	NHSY	OETP	RSLN	AYDN
5	EACA	BELE	EPRA	TAGN	NMXW	NSHO
6	YOIE	WITD	FTEO	TERD	TCHA	TIDN
7	FEET	EYGE	TANE	MAPS	IERC	NEEH
1	LLSEIPWLUYETTHACAIADESBDIBITREECOEHOTOEHORHUTOIE					
2	TMWMLKRIIPLLDYRNNLSAEOGRNHIHIECRPTEDTCERMSOWEIRN					
3	YETDVTRAHNELNHEESLOPIRMEAHNHUEFUENEHOENRNEEHDTGW					
4	SAO	BLRE	TAMA	WEHH	RBTF	YATC Q
5	AOR	ETSM	AIND	ITRN	AMRA	AUTE F
6	HEL	TIRO	DDSO	ETIS	LIHD	ABEF G
7	WHO	TTSS	MIEF	DCEE	MFGM	PDRA A

Fig. 1-3a,

Collapse the wave so that all the columns are filled as shown in Fig 1-3b.

	1	2	3	4
	123456789012345678901234567890123456789012345678			
1	LLSEIPWL	UYETTHACAIADESBDIBITREECOEHOTOEHORHUTOIE		
2	TMWMLKRIIPLLDYRNNLSAEOGRNHIHIE	CRPTEDTCERMSOWEIRN		
3	YETDVTRAHNELNHEESLOPIRMEAHNHUEFUENEHOENRNEEHDTGW			
4	SAOLLSDBLRERNTHTAMANHSYWEHHOETPRBTFRSLNYATCAYDNQ			
5	AOREACAETSMBELEAINDEPRAITRNTAGNAMRANMXWAUTENSHOF			
6	HELIOIETIROWITDDDSOFTEOETISTERDLIHDTCHAABEFTIDNG			
7	WHOFEETTSSEYGEMIEFTANEDCEEMAPSMFGMIERCPDRANEEHA			

Fig. 1-3b

For brevity sake, display only rows 25 through 48 of Fig 1-3b as 'W', shown in Fig 1-4.

	1	2
	123456789012345678901234	
1	LLSEIPWL	UYETTHACAIADESBD
2	TMWMLKRIIPLLDYRNNLSAEOGR	
3	YETDVTRAHNELNHEESLOPIRME	
4	SAOLLSDBLRERNTHTAMANHSY	
5	AOREACAETSMBELEAINDEPRAI	
6	HELIOIETIROWITDDDSOFTEOE	
7	WHOFEETTSSEYGEMIEFTANED	

Fig. 1-4, W

Take the rows of W and arrange them as follows:

R5:R1
R6:R2
R7:R3

and as shown below in W1, Fig 1-5:

	1	2		1	2
	1	2	3	4	5
	2	3	4	5	6
	3	4	5	6	7
	4	5	6	7	8
	5	6	7	8	9
	6	7	8	9	10
	7	8	9	10	11
	8	9	10	11	12
	9	10	11	12	13
	10	11	12	13	14
	11	12	13	14	15
	12	13	14	15	16
	13	14	15	16	17
	14	15	16	17	18
	15	16	17	18	19
	16	17	18	19	20
	17	18	19	20	21
	18	19	20	21	22
	19	20	21	22	23
	20	21	22	23	24
	21	22	23	24	25
	22	23	24	25	26
	23	24	25	26	27
	24	25	26	27	28
	25	26	27	28	29
	26	27	28	29	30
	27	28	29	30	31
	28	29	30	31	32
	29	30	31	32	33
	30	31	32	33	34
	31	32	33	34	35
	32	33	34	35	36
	33	34	35	36	37
	34	35	36	37	38
	35	36	37	38	39
	36	37	38	39	40
	37	38	39	40	41
	38	39	40	41	42
	39	40	41	42	43
	40	41	42	43	44
	41	42	43	44	45
	42	43	44	45	46
	43	44	45	46	47
	44	45	46	47	48
	45	46	47	48	49
	46	47	48	49	50
	47	48	49	50	51
	48	49	50	51	52
	49	50	51	52	53
	50	51	52	53	54
	51	52	53	54	55
	52	53	54	55	56
	53	54	55	56	57
	54	55	56	57	58
	55	56	57	58	59
	56	57	58	59	60
	57	58	59	60	61
	58	59	60	61	62
	59	60	61	62	63
	60	61	62	63	64
	61	62	63	64	65
	62	63	64	65	66
	63	64	65	66	67
	64	65	66	67	68
	65	66	67	68	69
	66	67	68	69	70
	67	68	69	70	71
	68	69	70	71	72
	69	70	71	72	73
	70	71	72	73	74
	71	72	73	74	75
	72	73	74	75	76
	73	74	75	76	77
	74	75	76	77	78
	75	76	77	78	79
	76	77	78	79	80
	77	78	79	80	81
	78	79	80	81	82
	79	80	81	82	83
	80	81	82	83	84
	81	82	83	84	85
	82	83	84	85	86
	83	84	85	86	87
	84	85	86	87	88
	85	86	87	88	89
	86	87	88	89	90
	87	88	89	90	91
	88	89	90	91	92
	89	90	91	92	93
	90	91	92	93	94
	91	92	93	94	95
	92	93	94	95	96
	93	94	95	96	97
	94	95	96	97	98
	95	96	97	98	99
	96	97	98	99	100
	97	98	99	100	101
	98	99	100	101	102
	99	100	101	102	103
	100	101	102	103	104
	101	102	103	104	105
	102	103	104	105	106
	103	104	105	106	107
	104	105	106	107	108
	105	106	107	108	109
	106	107	108	109	110
	107	108	109	110	111
	108	109	110	111	112
	109	110	111	112	113
	110	111	112	113	114
	111	112	113	114	115
	112	113	114	115	116
	113	114	115	116	117
	114	115	116	117	118
	115	116	117	118	119
	116	117	118	119	120
	117	118	119	120	121
	118	119	120	121	122
	119	120	121	122	123
	120	121	122	123	124
	121	122	123	124	125
	122	123	124	125	126
	123	124	125	126	127
	124	125	126	127	128
	125	126	127	128	129
	126	127	128	129	130
	127	128	129	130	131
	128	129	130	131	132
	129	130	131	132	133
	130	131	132	133	134
	131	132	133	134	135
	132	133	134	135	136
	133	134	135	136	137
	134	135	136	137	138
	135	136	137	138	139
	136	137	138	139	140
	137	138	139	140	141
	138	139	140	141	142
	139	140	141	142	143
	140	141	142	143	144
	141	142	143	144	145
	142	143	144	145	146
	143	144	145	146	147
	144	145	146	147	148
	145	146	147	148	149
	146	147	148	149	150
	147	148	149	150	151
	148	149	150	151	152
	149	150	151	152	153
	150	151	152	153	154
	151	152	153	154	155
	152	153	154	155	156
	153	154	155	156	157
	154	155	156	157	158
	155	156	157	158	159
	156	157	158	159	160
	157	158	159	160	161
	158	159	160	161	162
	159	160	161	162	163
	160	161	162	163	164
	161	162	163	164	165
	162	163	164	165	166
	163	164	165	166	167
	164	165	166	167	168
	165	166	167	168	169
	166	167	168	169	170
	167	168	169	170	171
	168	169	170	171	172
	169	170	171	172	173
	170	171	172	173	174
	171	172	173	174	175
	172	173	174	175	176
	173	174	175	176	177
	174	175	176	177	178
	175	176	177	178	179
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	177	178	179	180	181
	178	179	180	181	182
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	180	181	182	183	184
	181	182	183	184	185
	182	183	184	185	186
	183	184	185	186	187
	184	185	186	187	188
	185	186	187	188	189
	186	187	188	189	190
	187	188	189	190	191
	188	189	190	191	192
	189	190	191	192	193
	190	191	192	193	194
	191	192	193	194	195
	192	193	194	195	196
	193	194	195	196	197
	194	195	196	197	198
	195	196	197	198	199
	196	197	198	199	200
	197	198	199	200	201
	198	199	200	201	202
	199	200	201	202	203
	200	201	202	203	204
	201	202	203	204	205
	202	203	204	205	206
	203	204	205	206	207
	204	205	206	207	208
	205	206	207	208	209
	206	207	208	209	210
	207	208	209	210	211
	208	209	210	211	212
	209	210	211	212	213
	210	211	212	213	214
	211	212	213	214	215
	212	213	214	215	216
	213	214	215	216	217
	214	215	216	217	218
	215	216	217	218	219
	216	217	218	219	220
	217	218	219	220	221
	218	219	220	221	222
	219	220	221	222	223
	220	221	222	223	224
	221	222	223	224	225
	222	223	224	225	226
	223	224	225	226	227
	224	225	226	227	228
	225	226	227	228	229
	226	227	228	229	230
	227	228	229	230	231
	228	229	230	231	232
	229	230	231	232	233
	230	231	232	233	234
	231	232	233	234	235
	232	233	234	235	236
	233	234	235	236	237
	234	235	236	237	238
	235	236	237	238	239
	236	237	238	239	240
	237	238	239	240	241
	238	239	240	241	242
	239	240	241	242	243
	240	241	242	243	244
	241	242	243	244	245
	242	243	244	245	246
	243	244	245	246	247
	244	245	246	247	248
	245	246	247	248	249
	246	247	248	249	250
	247	248	249	250	251
	248	249	250	251	252
	249	250	251	252	253
	250	251	252	253	254
	251	252	253	254	255
	252	253	254	255	256
	253	254	255	256	257
	254	255	256	257	258
	255	256	257	258	259
	256	257	258	259	260
	257	258	259	260	261
	258	259	260	261	262
	259	260	261	262	263
	260	261	262	263	264
	261	262	263	264	265
	262	263	264	265	266
	263	264	265	266	267
	264	265	266	267	268
	265	266			

For ease, I have provided a python script that automates this in my folder in the Yahoo Kryptos Group Files under personal folders, or one could use one of the interactive keyed Vigenère solvers on the web like Rumkin's Keyed Vigenère Cipher at <http://rumkin.com/tools/cipher/vigenere-keyed.php>

Continuing: Arrange K4 in 6 rows so that they look like small "pyramids" or perhaps "A's" on their sides as shown in K4-H Fig 1-6. Also shown in Fig 1-6a is K4 arranged in it's pad configuration.

```

1      OBKRUOXOGHULBSO
2  LIFBBWFLRVQQPRNGKSSO
3      TWTQSQJSSEKZZW

4  ATJKLUDIAWINFB
5  NYPVTTMZFPKWGDKZXTJC
6  DIGKUHUAUEKCAR

```

Fig. 1-6, K4-H

```

          1          2      0          1          2
1234567890123456789012344123456789012345678901234
ATJKLUDIAWINFB      OBKRUOXOGHULBSO
NYPVTTMZFPKWGDKZXTJC  LIFBBWFLRVQQPRNGKSSO
DIGKUHUAUEKCAR      TWTQSQJSSEKZZW

```

Fig. 1-6a, K4-H

And apply K4 in Fig 1-6a over the negative space of the H in W1, Fig 1-5b, as shown in Fig 1-6b in an OTP decrypt.

```

          1          2      0          1          2
1234567890123456789012344123456789012345678901234
R5:R1  AOREACAETSMBELEAINDEPRAILLSEIPWL  UYETTHACAIADESBD
R6:R2  HELYOIETIROWITDDDSOFTEOETMW  MLKRIIPLLDYRNNLSAEOGR
R7:R3  WHOFEETTSSEYGEMIEFTANEDYETDVTRAH  NELNHEESLOPIRME

```

Fig. 1-6b

In other words:

DEC (UYETTHACAIADESB, OBKRUOXOGHULBSO) = DSIWLLLVSZHAWKW

DEC (HOFEETTSSEYGE, DIGKUHUAUEKCAR) = VDRIDDLPIOIAQJ

1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4

Y P O N B H Z E R O A B U W D S I W L L L V S Z H A W K W

B E R L I N H D E X P B C K J I M Z T O Z X F A N B C K D F M N D B Y A N U R M

V D R I D D L P I O I A Q J E F G R M O C K I S W D X O

1	DSIWLLLVSHAWKW	YPONBHZEROABUW
2	ZXFANBCKDFMNDBYANURM	BERLINHDEXPBCKJIMZTO
3	EFGRMOCKISWDXO	VDRIDDLPIOIAQJ

Don't worry, there are ways to tell if you have the right ones. The rows labeled with P1M are the rows of the correct letters where a misspelling occurs in the plaintext, as best I can guess. The rows labeled with P1 are the plaintext row from the OTP step. The rows labeled with W1 are the rows from the pad text and show the corresponding pad letter where the misspelling occurred in the plaintext.

		1		2			1		2
	123456789012345678901234					123456789012345678901234			
P1M 1		S		O		E		Z	
P1 R1		DSIWLLLVSHAWKW				YPONBHZEROABUW			
W1-14		UYETTHACIADESB				OREACAETSMBELE			
P1M 2		Q		I					S R
P1 R2		ZXFANBCKDFMNDBYANURM				BERLINHDEXPBCKJIMZTO			
W1-25		MLKRIIPLLDYRNNLSAEOG				ELYOIETIROWITDDDSOFT			
P1M 3		S		AR		D		E	S
P1-R3		EFGRMOCKISWDXO				VDRIDDLPIOIAQJ			
W1-36		NEINHEESLOPIRM				HOFEETTSSEYGE			

Fig. 1-9

Which gives us PT-H, our K4-H solution to the 97 characters as shown in Fig. 1-10

```

1      DSISLLLVSOHAWKW YEONBHZEROABUZ
2 ZXFANBCQDFINDBYANORM BERLINHDEXPBCKJIMSTR
3      EFGSMARKISWDXO  DDRIDDLEIOISQJ

```

Fig. 1-10, PT-H

Examining the misspellings, Fig. 1-11, we can find further messages. Read the misspelled letters across, left to right, by rows going down in such a way to get the following, PT-H:M1, as shown in Fig. 1-11.

SO EZ Q IOS RSA R DES

Fig. 1-11, PT-H:M1

Additionally if we use all three letters (the misspelling, plaintext, and pad letters) at each misspelling point we (maybe ???) can find another message:

SWT OZI EPR ZWE QKL IMY OUE SZO ROT SRN AOE RCE DVH EPT SAY

Fig. 1-12, PT-H:M2

Part 2: K4-O

Bismillahi Al Haqq il Ala, let us continue.

Carryout Transpositions Step:

As in part 1, start with W in Fig 1-4, but this time re-arrange the rows as follows:

R4:R3
R1:R6
R5:R7

shown below in W2, Fig 2-1, and draw the letter 'O' on it in the following way:

1 2 0 1 2
123456789012345678901234123456789012345678901234

R5:R3 AOREACAETSMBELEAINDEPRAIYETDVTRAHNEELNHEESLOPIRME
R1:R6 L LSEIPWLUYETTHACAIADIESBDHELYOIETIROWITDDDSOFTEOE
R4:R7 SAOLLSDBLRERNTHHTAMANHSYWWHOFEETTSSEYGEMIEFTANED

Fig. 2-1, W2

Continuing: Arrange K4 in as shown in K4-O, Fig 2-2a. Note that 'NYPVTTMZFPKWG' is repeated. Also shown in Fig 2-2b is K4 arranged in it's pad configuration.

O1 OBKRUOXOGHULBSOLIFBBW
O2 FLRVQQPRNGKSSOTWTQSJQ
O3 SSEKZZWATJKLUDIAWINFBN
O4a YPVTTMZFPKWD
O5 KZXTJCDIGKUHUAUEKCAR

O4b NYPVTTMZFPKWG

Fig. 2-2a, K4-O

1 2 0 1 2
123456789012345678901234123456789012345678901234

O1:O2 OBKRUOXOGHULBSOLIFBBWFLRVQQPRNGKSSOTWTQSJQ
O4a:b YPVTTMZFPKWD NYPVTTMZFPKWG
O3:O5 SSEKZZWATJKLUDIAWINFBNKZXTJCDIGKUHUAUEKCAR

Fig. 2-2b, K4-O

OTP, One Time Pad, Decryption Step:

As before in part 1, do the OTP decryption as follows. Note that the last two are actually 'Encryptions':

DEC (EACAETSMBELEA INDEPRAI, OBKRUOXOGHULBSOLIFBBW) =
QRLQDRMGOPTSRLFATCARB

DEC (LSEIPWLUYETH, YPVTTMZFPKWGD) = EWEIRUBLRINCV

DEC (OLLSDBLRERNTH TAMANHSYW, SSEKZZWATJKLUDIAWINFBN) =
RIQQILSSNIAGASBIJVOSSV

DEC (WHOFEETTTSESSEY GEMIEFT, KZXTJCDIGKUHUAUEKCAR) =
PENMOXSECQIPNQDNEXUW

ENC (YETDVTRAHNELNHEESLOPI, FLRVQQPRNGKSSOTWTQSJQ) =
HYOSJXTBASEWZNIBDEENC

ENC (WITDDDSOFTEOE, NYPVTTMZFPKWGD) = JLASHHXTXAEYX

The plain text is backwards, so arrange them as in P2 shown in Fig. 2-3.

1a BRACTAFLRSTPOGMRDQLRQ
2a VCNIRLBURIEWEY
3a VSSOVJIBSAGAINSSLIQQIRWUXENDQNPIQCESXOMNEP

1b CNEEDBINZWESABTXJSOYH
2b JLASHHXTXAEYX

Fig. 2-3, P2

Misspellings Step:

Guess the misspellings as follows in Fig 2-4. The rows labeled with P2M are the rows of the correct letters where a misspelling occurs in the plaintext, as best I can guess. The rows labeled with P2 are the plaintext row from the OTP step. The rows labeled with W are the row from the pad text and shows the corresponding pad letter where the misspelling occurred in the plaintext.

		1	2	1	2																				
		123456789012345678901234	12345678901234	12345678901234	1234																				
P2M 1		A	L	O	O	T	S	I																	
P2 R1		QRLQDRMGOPTSRLFATCARBHYOSJXTBASEWZNIBDEENC																							
W R5:R3		AOREACAETSMPLEA INDEPRAIYETDVTRAHNELNHEESLOPIRME																							
P2M 2		D		C	E			T	P	K															
P2 R2		EWEIRUBLRINCV												JLASHHXTXAEYX											
W R1:R6		LISEIPWLUYE THACAIADESBDHELYOIETIRO WITDDDSOFTEOE																							
P2M 3			R	I		M			O		E	HY		Q											
P2 R3		RIQQILSSNIAGASBIJVOSV PENMOXSECQIPNQDNEXUW																							
W R4:R7		SAOLLSDBLRERNTH TAMANHSYWWHOFETTSSEY GEMIEFTANED																							

Fig. 2-4

Which gives us PT-O, our K4-O solutions as shown in Fig 2-5.

```

BRACTOFLOSTLOGARDQLRQ
          VENCRLBURIEDE
VSSOVJIMSAGAINISLRQQIR  QUXENDQHPIECESXOONEP

```

Fig. 2-5a, PT-Oa

```

INEEDBINSWESTBTXJSOYH

```

Fig. 2-5b, PT-Ob

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JLASTHPTXKEY

```

Fig. 2-5c, PT-Oc

Examining the misspellings we can find further messages. Read off the misspelled letters of Fig 2-4 starting by going up on odd rows, down on even, right to left for the right set of 24 columns and continuing then to the left set of 24 columns by going down on odd rows, up on even, to get the following, PT-O:M, as shown in Fig 2-6.

```

KIQ P TY H SET O - O MOLE CIA RD

```

Fig. 2-6, PT-O:M

Perhaps one can find, using the misspelling, plaintext, and pad letters at each misspelling point, other messages as in part one or as in other solutions. Maybe.

Part 3: Y

Bismillahi Ar Rahmani Raheem, let us continue.

Carryout Transpositions Step:

As in part 1, start with W in Fig 1-4, but this time re-arrange the rows as shown below in W3, Fig 3-1, and draw an 'Awkward' Y on it in the following way:

	1	2
	123456789012345678901234	
R5	AOREACAETSMBEI	EAINDEPRAI
R2	TMWMLKRIIPLLDYRNNI	SAEOGR
R4	SAOLLSDBLRERNTH	TAMANHSYW
R3	YETDVTRAHNELNHEESLOPIRME	
R7	WHOFEETTTTSSEYGEMIEFTANED	
R1	LLSEIPWLUYETTHACAIADIESBD	

Fig. 3-1, W3

Continuing: Arrange K4 in as shown in K4-Y, Fig 3-2. Note that 'NYPVTTMZFPKWG' is not used. It was already used in K4-O. Note also that row Y3 is made up of two parts, RNGKSSO and UEKCAR. As an aside, see that when reversed and put together they form OSSKGNRRACKEU - if we replace the NR and U with a BC and a D we get OSSKGBCRACKED or OSS KGB CRACKED. In K4-Y and W3 there seems there might be some interesting stenography. Also shown in Fig 3-2b is K4 arranged in it's pad configuration.

Y1	OBKRUOXOGHULBS
Y2	OLIFBBWFLRVQQP
Y3a	RNGKSSO
Y4	TWTQSJQSSEKZZW
Y5	ATJKLUDIAWINFB
Y6	NYPVTTMZFPKWGD
Y7	KZXTJCDIGKUHUA
Y3b	UEKCAR

Row Y3: RNGKSSOUEKCAR

Fig. 3-2a, K4-Y

Y5	TJKLUDIAWINFB
Y1	OBKRUOXOGHULB
Y3	RNGKSSOU EKCAR
Y4	WTQ SJQSSEKZZW
Y7	ZXTJCDIGKU HUA
Y2	OLIFBBWFLRVQQ

Fig. 3-2b, K4-Y

OTP, One Time Pad Step:

As Before do the OTP decrypt step on each of the parts:

ENC (	OREACAETSMBEL	,	TJKLUDIAWINFB	) =	CLEXTLKEPARWZ
ENC (	KRIIPLLDYRNNL	,	OBKRUOXOGHULB	) =	OCIJXVIII IHDZ
ENC (	BLRERNTH TAMAN	,	UEKCARRNGKSSO	) =	CDHEAZCC IARHQ
DEC (	LNHEESLOPIRME	,	WTQ SJQSSEKZZW	) =	SESUOHIRBEXAF
DEC (	FEETTTSS EYGEM	,	ZXTJCDIGKU HUA	) =	GGNFOSCA INRDI
DEC (	LSEIPWLUYETTH	,	OLIFBBWFLRVQQ	) =	HETWOESLIJMJS

Misspellings Step:

Guess the misspellings as follows in Fig 3-4. The rows labeled with P3M are the rows of the correct letters where a misspelling occurs in the plaintext, as best I can guess. The rows labeled with P3 are the plaintext row from the OTP step. The rows labeled with W are the row from the pad text and shows the corresponding pad letter where the misspelling occurred in the plaintext.

			1				2																	
	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	1	2	3	4
P3 R1	CLEXTLKEPARWZ																							
W R5	OREACAETSMBEL																							
P3 R2	OCIJXVIII IHDZ																							
W R2	KRIIPLLDYRNNL																							
P3 R2	CDHEAZCC IARHQ																							
W R4	BLRERNTH TAMAN																							

Fig. 3-4a

										1										2													
										1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	1	2	3	4
P3M	3																																
P3	R4																																
W	R3																																

P3M	4																																
P3	R5																																
W	R7																																

P3M	5																																
P3	R6																																
W	R1																																

Fig. 3-4b

As far as the first part of the 'Y', Fig 3-4a, I am uncertain of the misspellings. I believe that there is more than one solution. Please find your own. I will give one possible plaintext, PT-Ya:

C TEXT IKE PART TO CIA X V IIIIH D AWD HEAD DCIA R HQ

Fig. 3-5a, PT-Ya

Which gives us PT-Yb 3.1, our K4 part 3.1 solution as shown in Fig 3-5.

SOS U OHB R B EXACT GN POS CT IN RD I H V TWO IS LIOM JS

Fig. 3-5b, PT-Yb

There might be further messages in the misspellings, as in the other solutions. I have not found any that I am well pleased with.

Interpretations:

What follows are my best guesses to date - certainly not everything will be correct.
CQD! SOS!

Plaintexts and others:

PT-H:

D SIS LLL VS O-H-AWK WYE ON BH ZERO ABUZZ X
FAN B CQD FIND BY ANORM BERLIN HD EXP BCK JIM S
T REF, GS MARK IS WD X ODD RIDDLE IO IS Q J

PT-Oa:

BR ACT OF LOST LOG A RD Q
LRQ V ENCR L BURIED E VS SOV
JIM S AGAIN IS LRQ Q
I RQ U, X, ENDY, H PIECES X
O ONE P

PT-Ob:

I NEED BINS WEST B TX JS O,Y,H

PT-Oc:

J LAST H PT X KEY X

PT-Ya:

C TEXT IKE PART TO CIA X, V IIIIH, D AWD HEAD DCIA R HQ

PT-Yb:

SOS U, OHB R B, EXACT GN POS CT IN RD I,
H V TWO IS LIOM, JS

PT-H:M1

SO EZ Q IOS RSA R DES

PT-H:M2

S W TO ZIE PRZ WEQK L IM YUES Z O ROTS R NAOE RCED V HEP T SAY

PT-O:M

KIQ P TY H SET O - O MOLE CIA RD

PT-H

D¹ SIS² LLL VS O-H-AWKward WYE³ ON BH ZERO⁴ ABUZZ⁵ X
FAN of B CQD⁶ - FIND BY AbNORMAl BERLIN HeaD,EXPeCt BaCK - Jim S.
T is REFeRence⁷
GS MARK IS WD⁸ X

ODD RIDDLE - 10 IS?⁹ Jim

1. D is for Dulles, the DCI at the time, and in general the CIA.
2. SIS: www.sis.gov.uk , The British Secret Intelligence Service, aka MI6
3. LLL vs O-H-WYE: a riddle of some sort. LLL (maybe Lies) from the CIA and MI6 verses OH-WYE. OH might be a code for Kim Philby and the Cambridge 5 (V - five). Or it might be "OHW-YE" as in 'Who Ye' as part of John 8:32 'And Ye Shall Know ...' which is carved into a wall on the OHB (Old Headquarters Building) in Langley. Or both. Also it could read 'Trial' or 'Try L'. Therefore:

ENC (L, OHWYE) = V OHNY

Maybe that means to 'V' 'OHNY' so (remember 'Level Two'):

ENC (V, OHNY) = RD I X

Maybe that means that we should 'X Riddle One' i.e. encrypt K4-H with an X.

Also, WYE, this the word for the letter 'Y' - which could also mean 'Why'. Or it could mean 'W Ye' as in 'W's Ye' - Allen Welsh Dulles's Ye - again a reference to John 8:32.

4. On BH Zero: OHB written backwards (Old Headquarters Building, Langley). Or BH as in 'Berlin Head Zero' (might be a bug - and should read 'Hero' ??? depending on one's point of view?). Berlin Head would be a reference to Allen Dulles, again, who was head of the OSS in Berlin just after WWII.

5. ABUZZ - 'A BUG Z'? See the misspelling for K4-H.

6. FAN B CDQ: A Fan of 'B', maybe 'B' stands for the OHB? CQD is an old form of SOS. It could also be meant to be read 'Seek D'.

7. T (for Truth) is our reference point referring to 'T IS YOUR POSITION' in Morse code next to the loadstone. The compass is pointing SW - (maybe SW is short for the SIS and W (Allen Dulles)???)

8. GS MARK IS WD X - on the USGS Marker that was in front of the NHB, New Headquarters Building. On it is written the word (letter) 'X' (or a cross, depending on your point of view)

9. Odd riddle: yes this is. Ten is?

ENC (V, ENDY) = AISX

Ten is A!

PT-H:M1

SO Easy? aS RSA oR DES

Remember A is IO (that also might mean that A is 2, as IO is 2 in binary). And since we have 'Level Two'

ENC (AISX, SOEZQIOSRSARDES) = GQLWRTETBUGZLKF - GQ L, WRTE T, BUG Z, LKF

Perhaps: Go to L and Write T; there is a Bug Z; LKF ???

PT-H:M2

S W TO ZIE PRZ WEQK L IM YOUES Z O ROTS R NAOE RCED V HEP T SAY

Maybe something:

SIS, Allen W Dulles to Lie (or SW compass heading - referring to T to Die) , Mr President (it's) a weak lie.
I'm Yours, O Rots our name, Received V Help Say the Truth ??? - really I don't know.

PT-Oa:

BRother¹ ACT OF LOST LOG A Riddle? Location Request:

V ENCRypted L BURIED E VS SOVIets

JIM S. AGAIN IS (this) a Location Request? I require you, a X,
the ENDY, and H PIECES² X O one piece.³

1. Brother meaning the British, or just 'British'
2. 'ENDY' is the ENDYAHROHN piece, 'H' meaning our PT-H
3. A hint that the mask for this part forms an O, or perhaps 'O is a one Philby'?

PT-Ob:

I NEED BIN'S¹ WEST Berlin Text, Jim S. O-Y-H²

1. Binary or it might refer to JFK's "Ich bin Ein Berliner" speech.
2. O-Y-H a clue to there being 3 plaintexts

PT-Oc:

Jim - LAST H Plaintext X KEY X

The last part of PT-H is key meaning the riddle of IO is ? A is X! And that is a Key.
Also a hint as what to do with the PT-H plaintext.

PT-O:M

KIQ P TY H SET O - O MOLE CIA RD

Kim Philby TY H SET(the Cambridge V) O, O MOLE CIA Riddle (or CIA is our D)

PT-Ya:

I'm really guessing here, multiple interpretations may exist:

C TEXT IKE PART TO CIA X, V IIIIH, D AWD HEAD DCIA R HQ

See text from Ike - the part to the CIA X

V is IIIIH¹, D² is AWD, Allen W. Dulles,
HEAD DCI, CIA³ is our Headquarters

1. the Cambridge 5 spies, H is Philby?

2. or HD - repeating the H from the last part.

3. DCI - CIA, repeating the letters CI

PT-Yb:

SOS from you - OHB is our 'B'

the EXACT ground POSition in ciphertext in Riddle One, H X¹, is L-IO-M²,
Jim Sanborn

1. Two V's is a X ($5 + 5 = 10$, $V + V = X$)

2. This is almost the same as LOOMIS, which was reported to be written on the USGS marker.

Other points:

Whom is K4-H from and to who is it addressed?

Consider, from W1, Fig 1-5, take the 'LWPIESLL' and reverse it to get 'LLSEIPWL'. Take the last 8 letters of K4, 'UAUEKCAR' and use 'LLSEIPWL' as the key to decrypt 'UAUEKCAR'

`DEC(LLSEIPWL, UAUEKCAR) = TJIKESDD`

Or 'TO IKES DD'; i.e. "To Ike's Deputy Director" with the J for Jim as a misspelling and a signature. Now when Eisenhower (Ike) became president, Allen Welsh Dulles was the DD, Deputy Director, of the CIA. Shortly after, Ike promoted him to DCI, Director of Central Intelligence, the head of the CIA.

Also, note from W1, on the other side of the H - it goes "AIN". Putting the two together we get 'LLSEIPWL-AIN'. If the 'W' is replaced by a 'S' it becomes LLSEIPSLAIN. So one way, from the 'S' covered up by 'W' it reads 'SPIES LL', the other direction from the 'S' it reads 'SLAIN'. Maybe it should read 'Slain Spies'. Maybe 'W' is for Allen W. Dulles

So maybe K4 is from SPIES (possibly Slain Spies) to Allen W. Dulles.

In addition, Allen Dulles was head of the OSS Berlin station in the days just after WWII and he oversaw the building of the OHB, the Original Headquarters Building, of the CIA at Langley. (The CIA refers to it as the OHB - just see their website. Kryptos was commissioned as part of the NHB, the New Headquarters Building)

More:

One might be able to turn the 'Y' mask, K4-Y, into an upside down 'e' by using a piece found next to the 'WYE' at the left edge of W3, R3 and R7 with the first 7 letters of Y4 from K4-Y.

DEC (TDVTRAH, TWTQ SJQ) = KGBJOCS

Or 'KGB DOCS' - KGB documents, with the J for Jim as a misspelling and a signature.

Salam and Happy Discoveries and Adventures,
Ibn Şifre

SUMHGGJKKIPETCTWGTGGPIBZJEJNJKKQRXWCCYXFSBWMJAZUMC